

D 130227

(Pages : 3)

Name.....

Reg. No.....

**FIFTH SEMESTER CBCSS-UG DEGREE EXAMINATION
NOVEMBER 2025**

(2020 Syllabus)

Mathematics

MTS 5B 06—BASIC ANALYSIS

Time : Two Hours and a Half

Maximum : 80 Marks

Section A*Short Answer Type**All questions can be answered.**Each question carries 2 Marks. (Ceiling 25 Marks)*

1. Let $a \in \mathbb{R}$. Prove that $a \cdot 0 = 0$
2. Let $a \in \mathbb{R}$. with $a \neq 0$. Prove that $a^2 > 0$.
3. Let x, y belongs to the ϵ -neighbourhood of a, b . Prove that $x + y$ belongs to the 2ϵ -neighbourhood of $a + b$.
4. Prove that an upper bound of u of a non-empty set S in $\mathbb{R}$ is the supremum of S if and only if for every $\epsilon > 0$ there exist an $s_\epsilon \in S$ such that $u - \epsilon < s_\epsilon$.
5. Let S be a non-empty subset of $\mathbb{R}$ that is bounded above and let $a \in \mathbb{R}$. Prove that $\sup(a + s) = a + \sup S$.
6. Consider the sequences $X = (2, 4, 6, \dots, 2n, \dots)$ and $Y = \left(\frac{1}{1}, \frac{1}{2}, \frac{1}{8}, \dots, \frac{1}{n}, \dots\right)$. Find XY and X/Y .
7. Let (x_n) be a sequence of real numbers that converges to x . Prove that $(|x_n|)$ converges to $|x|$.
8. Let (x_n) be a sequence of real numbers that converges to x . Prove that any subsequence (x_{n_k}) of (x_n) also converges to x .

Turn over

9. Whether the sequence $\left(1+(-1)^n\right)$ is a Cauchy sequence ? Justify your answer.
10. Let $c > 1$, find the limit $\lim(c^n)$.
11. Find the modulus of the complex number $\frac{2i}{3-4i}$.
12. Find the polar form of the complex number $-\sqrt{3}-i$.
13. Express the number $3i^5 - i^4 + 7i^3 - 10i^2 - 9$ in the form of $a + ib$.
14. Sketch the graph of $Re(z) = 5$ in the complex plane.
15. Find the real and imaginary parts u and v of the complex function $f(z) = z^3 - 2z + 6$.

(Ceiling 25 marks)

Section B

Paragraph / Problem Type.
All questions can be answered.
Each question carry 5 marks.
(Ceiling 35 marks).

16. Let A be any set. Prove that there is no surjection of A onto the set $\mathcal{P}(A)$ of all subsets of A .
17. Let $a, b \in \mathbb{R}$. Prove that (i) $|ab| = |a||b|$. (ii) $|a+b| \leq |a| + |b|$.
18. Let $x, y \in \mathbb{R}$ with $x < y$. Prove that there exist an irrational number z such that $x < z < y$.
19. Let $X = (x_n)$ be a sequence of real numbers that converges to x and $Z = (z_n)$ be a sequence of non-zero real numbers that converges to z with $z \neq 0$. Prove that X/Z converges to x/z .
20. Let (x_n) be a bounded increasing sequence. Prove that $\lim(x_n) = \sup\{x_n : n \in \mathbb{N}\}$.

21. Find $\operatorname{Re}(z)$ and $\operatorname{Im}(z)$ where $z = \left(\frac{i}{3-i}\right)\left(\frac{1}{2+3i}\right)$.
22. Describe the set of points z in the complex plane satisfy $|z-i| = |z-1|$.
23. Prove that the union of an arbitrary collection of open subsets in $\mathbb{R}$ is open.

(Ceiling 35 marks)

Section C*Essay type.**Answer any **two** of the following questions.**Each question carries 10 marks.*

24. Prove that the set $\mathbb{R}$ of all real numbers is uncountable.
25. State and prove Nested Intervals Property.
26. Let $X = (x_n)$ be a sequence of real numbers that converges to x and suppose that $x_n \geq 0$. Prove that the sequence $X = (\sqrt{x_n})$ of positive square roots converges to $\sqrt{x}$.
27. Find the image of the vertical line $x = 1$ under the complex mapping $w = z^2$.

(2 × 10 = 20 marks)